

Abstract

Product launches involve significant uncertainty due to unpredictable customer perception and rapidly changing market dynamics. Traditional evaluation approaches such as surveys and post-launch analytics fail to proactively assess risk before deployment. While Generative AI tools have gained popularity, most applications focus on content generation rather than simulation-driven decision support.

This project proposes **DryRun AI**, a GAN-centric pre-launch simulation platform designed to evaluate product and marketing strategies before market release. The core of the system is a Generative Adversarial Network (GAN) framework that generates realistic synthetic product visuals and controlled variations across attributes such as color, texture, layout, and presentation style. By exploring the latent space of GAN models, the system produces multiple optimized visual alternatives that simulate potential real-world representations of a product.

These synthetic variants are evaluated under multiple launch scenarios using rule-based simulation and LLM-driven persona reasoning to approximate audience reactions. The platform integrates optimization techniques such as latent space manipulation, hyperparameter tuning, scenario weighting, and threshold-based risk scoring to identify configurations that maximize engagement likelihood while minimizing predicted risk.

Unlike traditional marketing tools, DryRun AI positions GANs not merely as content generators but as a decision-support engine for strategic simulation. The system demonstrates how generative modeling combined with optimization-driven evaluation can reduce launch uncertainty, improve strategic planning, and provide measurable pre-launch insights.

1. INTRODUCTION

1.1 Background

Generative Adversarial Networks (GANs) have demonstrated remarkable capability in producing highly realistic synthetic images across domains such as fashion, product design, and visual media. However, their application has largely remained within creative content generation. In parallel, organizations continue to struggle with uncertain product launches due to limited pre-launch evaluation mechanisms. This project explores the novel application of GANs as a simulation engine rather than merely a generative tool.

1.2 Motivation

Product failure often arises from poor visual positioning, misaligned messaging, or incorrect pricing strategies. Traditional validation methods are expensive and biased. There is a need for a

system that can generate controlled variations of a product and evaluate them under multiple hypothetical scenarios. By leveraging GAN latent space exploration and optimization techniques, it becomes possible to identify product representations that maximize predicted engagement before launch.

1.3 Objectives

- To design a GAN-centric pre-launch simulation framework
- To generate controlled synthetic product variations using GAN latent manipulation
- To simulate audience perception under multiple configurable scenarios
- To optimize launch parameters using scenario-based evaluation metrics
- To minimize predicted risk and maximize engagement likelihood

1.4 Application

The system can be applied in fashion, consumer products, SaaS feature launches, and digital marketing campaigns where visual perception and positioning significantly influence adoption. It supports experimentation with visual style, pricing, and messaging prior to real-world deployment.

2. PROBLEM DEFINITION

Current product evaluation systems lack a generative simulation mechanism capable of exploring multiple realistic product representations before launch. Existing approaches either create content or analyze outcomes after deployment, failing to proactively reduce risk. There is a need for a GAN-driven simulation framework that can generate diverse yet controlled visual variations and evaluate them under multiple launch scenarios. Additionally, optimization mechanisms must be integrated to identify parameter configurations that maximize predicted engagement and minimize risk. Such a system would transform GANs from creative tools into strategic decision-support engines for pre-launch evaluation.

Month	Activity
February	Problem Statement Definition & Scope Finalization
March	Requirement Analysis & Initial Feasibility Study
April	Literature Review on GANs, Optimization Techniques & Simulation Systems
May	System Architecture Design (GAN Module, Scenario Engine, Backend Design)
June	Detailed System Design & Dataset Preparation
July	Implementation – GAN Model Integration & Backend Development
August	Implementation – Frontend Development & Scenario Engine Integration
September	Testing, Debugging, Performance Evaluation & Optimization
October	Final Deployment, Documentation & Project Submission